Buck Converter Failure Equation Analysis

Battery Management System for Electric Scooter | ECE 445 Team 35 | Technical Design Note

Design Context

The first master-board power-tree design used the TPS54308 buck converter to step the battery pack voltage down to 12 V. The converter was intended to operate from a lithium-ion pack input range of approximately 18 V to 25.2 V. The initial design used a 4.7 uH inductor at a switching frequency of 500 kHz. During testing, this value produced excessive inductor ripple current, which pushed the converter toward discontinuous conduction mode and degraded the stability of the 12 V rail.

Parameter	Value	Role in Analysis
V_in,max	25.2 V	Worst-case fully charged 6S pack voltage.
V_out	12 V	Regulated buck output rail.
f_s	500 kHz	Buck converter switching frequency.
L_initial	4.7 uH	Original inductor value that caused high ripple.
L_revised	10 uH	Corrected inductor value used to reduce ripple.
I_load,nom	2 A	Representative nominal load current for conduction-mode check.

Duty Cycle

For an ideal buck converter operating in continuous conduction mode, duty cycle is approximately the output voltage divided by the input voltage.

$$D = V_{out} / V_{in}$$

$$D = 12\ V / 25.2\ V = 0.476$$

At the maximum pack voltage, the converter switch is on for only about 47.6% of each switching cycle. This duty cycle is used in the inductor ripple-current calculation.

Initial Inductor Ripple Current

The peak-to-peak inductor ripple current for a buck converter is estimated by:

$$\Delta I_L = [V_{out} * (1 - D)] / [L * f_s]$$

Using the initial 4.7 uH inductor:

$$\Delta I_L = [12\ V * (1 - 0.476)] / [(4.7 \times 10^{-6}\ H) * (500 \times 10^{3}\ Hz)]$$

$$\Delta I_L = 2.67\ A \text{ peak-to-peak}$$

This ripple current is large compared with the expected load current. A large ripple current causes higher output ripple, increased RMS current stress, reduced regulation quality, and a greater chance of discontinuous conduction.

Discontinuous Conduction Mode Check

A buck converter enters discontinuous conduction mode when the inductor current reaches zero before the next switching cycle begins. A simple boundary check compares average inductor current to half the ripple current.

DCM risk occurs when $I_{L,avg} < \Delta I_L / 2$

$$\Delta I_L / 2 = 2.67\ A / 2 = 1.335\ A$$

For a light or moderate load, the inductor valley current is:

$$I_{L,valley} = I_{L,avg} - \Delta I_L / 2$$

If $I_{L,avg} = 1\ A$, then $I_{L,valley} = 1\ A - 1.335\ A = -0.335\ A$

A negative calculated valley current means the physical inductor current would hit zero, so the converter cannot remain in continuous conduction. Even near a 2 A nominal load, the margin is poor because the ripple current is a large fraction of the average current.

Output Ripple Impact

The output capacitor must absorb inductor ripple current. A first-order approximation for capacitive output ripple is:

Delta V_out approximately equals Delta I_L / (8 * f_s * C_out)

Because Delta V_out is directly proportional to Delta I_L, reducing inductor ripple current directly reduces output ripple. The initial 2.67 A ripple therefore increased output ripple and disturbed downstream 5 V and 3.3 V regulation.

Effect of Excessive Ripple	Impact on Board Operation
High inductor ripple	Large current swing through inductor, switch, diode/synchronous FET, and output capacitor.
DCM operation	Changes converter dynamics, reduces load regulation quality, and can make the output less predictable.
Higher 12 V ripple	Adds disturbance before the LDO rails and can affect sensitive analog measurement paths.
Higher RMS stress	Raises thermal and electrical stress on converter components.

Corrected 10 uH Inductor Calculation

The revised design increased the inductor to 10 uH while keeping the same voltage and switching-frequency assumptions.

Delta I_L = [12 V * (1 - 0.476)] / [(10 x 10^-6 H) * (500 x 10^3 Hz)]

Delta I_L = 1.26 A peak-to-peak

The half-ripple value becomes:

Delta I_L / 2 = 1.26 A / 2 = 0.63 A

This gives much better continuous-conduction margin. For example, at a 2 A load:

I_L,valley = 2 A - 0.63 A = 1.37 A

Since the valley current remains above zero, the converter stays in continuous conduction at this operating point. This improves regulation behavior and reduces ripple on the 12 V rail.

Summary of Failure Mechanism

- The 4.7 uH inductor produced approximately 2.67 A peak-to-peak ripple at 25.2 V input, 12 V output, and 500 kHz switching.
- The ripple was too large relative to the expected load current, so the converter was pushed toward discontinuous conduction mode.
- Discontinuous conduction and high ripple increased output voltage ripple, reduced regulation quality, and stressed downstream rails.
- Increasing the inductor to 10 uH reduced ripple to approximately 1.26 A peak-to-peak, improving continuous-conduction operation and stabilizing the 12 V rail.